

## EXPERIENCE

- Rural Studies Institute, GCSU (Sponsored by Georgia Tech PIN Program)** Milledgeville, GA  
*Front-End Developer & Data Visualization Intern* May 2025 – Aug 2025
- Designed and deployed interactive, data-driven maps for the Black Belt Regional Database using **JavaScript**, **Leaflet.js**, and **WordPress**.
  - Developed custom Python scraping scripts to aggregate socio-economic datasets from US Census (ACS) and USDA APIs/web portals, automating the ingestion of **10k+** data points and reducing manual data entry by **40%**.
  - Automated data pipelines converting spreadsheets to **GeoJSON**, reducing integration time by **30%**.
  - Collaborated with researchers and policymakers to launch a public-facing platform serving **500+** monthly users.
  - Presented solutions at the *Emerging Innovators Celebration*, enhancing project visibility.
  - Project Link:** gcsu.edu/ruralstudies/understanding-black-belt-region.

## TECHNICAL SKILLS

- Languages:** JavaScript/TypeScript, HTML/CSS, SQL, Python
- Frontend:** React.js, Next.js, Tailwind CSS, Fabric.js, React Beautiful DnD, Recoil, Zustand, Vite
- Backend:** Node.js, Express.js, RESTful APIs, Microservices, WebSockets, Socket.io
- Databases:** MongoDB, PostgreSQL, Prisma
- Testing & Quality:** Jest, Vitest, React Testing Library, Unit Testing, Test-Driven Development(TDD)
- Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS), Docker, Linux, CI/CD (GitHub Actions), Vercel, Render
- Developer Tools:** Git, Postman, UML, Zod, Bcrypt
- Soft Skills:** Agile/Scrum, Code Reviews, Technical Documentation, Cross-functional Collaboration

## PROJECTS

- TweenGen | React.js, Fabric.js, Recoil** [Github](#) | [Demo](#)
- Architected a lightweight animation engine with optimized easing algorithms, reducing the final bundle size by **15%** compared to standard external animation libraries.
  - Reduced canvas re-render cycles by **40%** through memoization and selective state updates in Recoil, maintaining a consistent **60 FPS** during complex multi-object animations.
  - Built drag-and-drop timeline UI with GSAP-powered live preview and code export system generating production-ready **HTML/CSS/JavaScript**. Hosted on Vercel.
- Real-Time Collaborative Task Board | React, Node.js, Socket.io, MongoDB** [Github](#) | [Demo](#)
- Built real-time collaboration system using **WebSockets**, enabling instant task updates across 50+ concurrent users with **<100ms latency**.
  - Designed **RESTful API** with **JWT** authentication, **Zod** validation, **Bcrypt** password hashing, and role-based access control
  - Improved data consistency by implementing a conflict resolution strategy for concurrent edits, reducing state synchronization errors by **15%** during peak user activity.
  - Deployed on **Vercel** (frontend) and **Render** (backend) with automated CI/CD and health monitoring.

## EDUCATION

- The University of Georgia** Athens, GA  
*Master of Science in Computer Science* Expected May 2026
- Coursework: Software Engineering, Database Management, Algorithms, Computer Networks, Data Science, Advanced Representation Learning
- Sathyabama Institute of Science and Technology** Chennai, India  
*Bachelor of Engineering in Computer Science* Jun 2024

## CERTIFICATIONS

- AWS Certified Developer - Associate - [link](#)
- AWS Certified Cloud Practitioner - [link](#)